Mathematical Analysis of Stochastic Perturbations in Astrocytic Metabolic Dynamics

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Abstract

We investigate how different stochastic perturbation schemes affect calcium-ATP coupling in astrocytes, with implications for understanding neurodegeneration in Parkinson's disease. Using a 30-dimensional ODE system extended to stochastic differential equations, we compare additive, multiplicative, state-dependent, and jump noise formulations. Our analysis reveals three principal findings: (1) the system exhibits hierarchical noise filtering, with ER calcium absorbing perturbations while ATP:ADP ratio remains stable; (2) inter-spike interval statistics follow Gamma rather than exponential distributions, indicating multi-step oscillation mechanisms; (3) a bifurcation at $IP_3 \approx 1.4\ \mu M$ separates oscillatory dynamics from metabolic collapse, where state-dependent noise provides stochastic rescue while additive noise permits cell death. These results connect stochastic dynamical systems theory to cellular pathophysiology.

1 Introduction

1.1 Parkinson's Disease and Calcium Signaling

Parkinson's disease (PD) is a progressive neurodegenerative disorder characterized by selective loss of dopaminergic neurons in the substantia nigra pars compacta. Clinical motor symptoms—tremor, rigidity, and bradykinesia—manifest after approximately 60–80% neuronal loss [1], indicating prolonged subclinical dysfunction preceding diagnosis.

At the molecular level, PD pathogenesis involves three interconnected mechanisms. First, the protein α-synuclein misfolds and aggregates into Lewy bodies, disrupting intracellular trafficking and organelle function [2]. Second, mitochondrial dysfunction, particularly Complex I deficiency, reduces ATP production and increases reactive oxygen species (ROS) generation [3]. Third, impaired mitophagy—the selective autophagy of damaged mitochondria—fails to remove dysfunctional organelles, particularly in cases involving PINK1 and Parkin mutations [4]. These processes form a pathogenic feedback loop: mitochondrial damage generates oxidative stress, which damages proteins and DNA, further compromising mitochondrial function.

Calcium ions (Ca^{2+}) serve as second messengers linking cellular activity to metabolic responses. Astrocytes respond to neuronal signals by releasing Ca^{2+} from endoplasmic reticulum (ER) stores via inositol 1,4,5-trisphosphate receptors (IP_3Rs), triggering metabolic cascades that increase ATP production [5]. This calcium-metabolism coupling is relevant to PD for several reasons. First, dopaminergic neurons have high energy demands due to extensive axonal arborization and autonomous pacemaking activity, making them dependent on astrocytic metabolic support [6]. Second, mitochondrial dysfunction directly impairs calcium handling since mitochondria both buffer cytosolic calcium and require calcium for optimal ATP synthesis [7]. Third, oxidative stress damages IP_3Rs and SERCA pumps, creating additional pathogenic feedback [8].

1.2 Calcium Dynamics in Astrocytes

This study investigates how stochastic perturbations to IP_3R-mediated calcium flux propagate through the metabolic network. The framework builds on the astrocyte-neuron lactate shuttle hypothesis, which proposes that astrocytes convert glucose to lactate and deliver it to neurons in response to calcium-mediated signals [9], as illustrated in Figure 1.

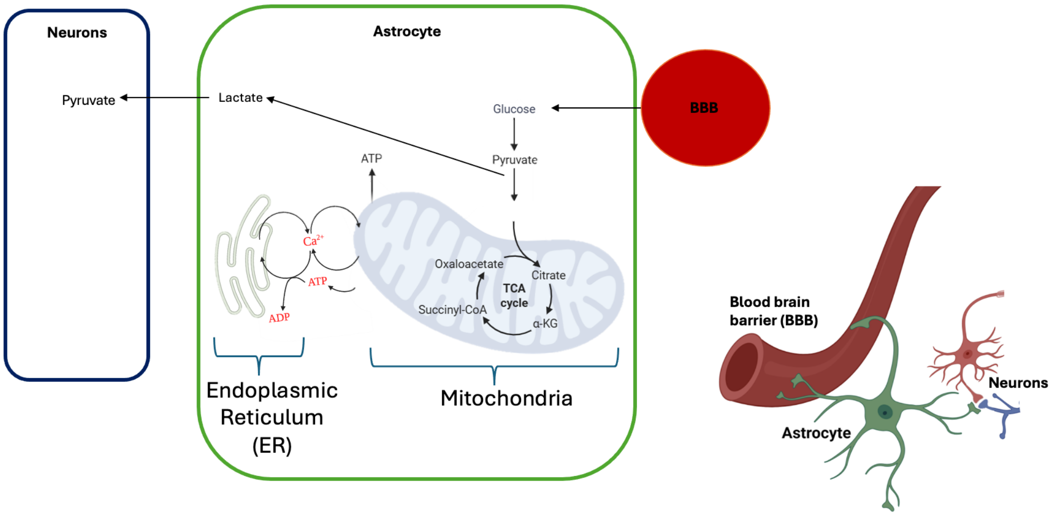


Figure 1: Schematic of astrocyte-neuron metabolic coupling showing glucose uptake, glycolytic processing, and lactate shuttle to neurons.

The ER maintains calcium concentrations approximately 10^4-fold higher than the cytosol ($\sim$500 μM vs $\sim$0.1 μM). IP_3Rs release ER calcium into the cytosol upon IP_3 binding, while SERCA pumps restore ER stores at the cost of ATP hydrolysis. This creates oscillatory dynamics: calcium release depletes ER stores, triggering energy-consuming pumps that reload the ER, producing rhythmic fluctuations in both calcium and ATP concentrations.

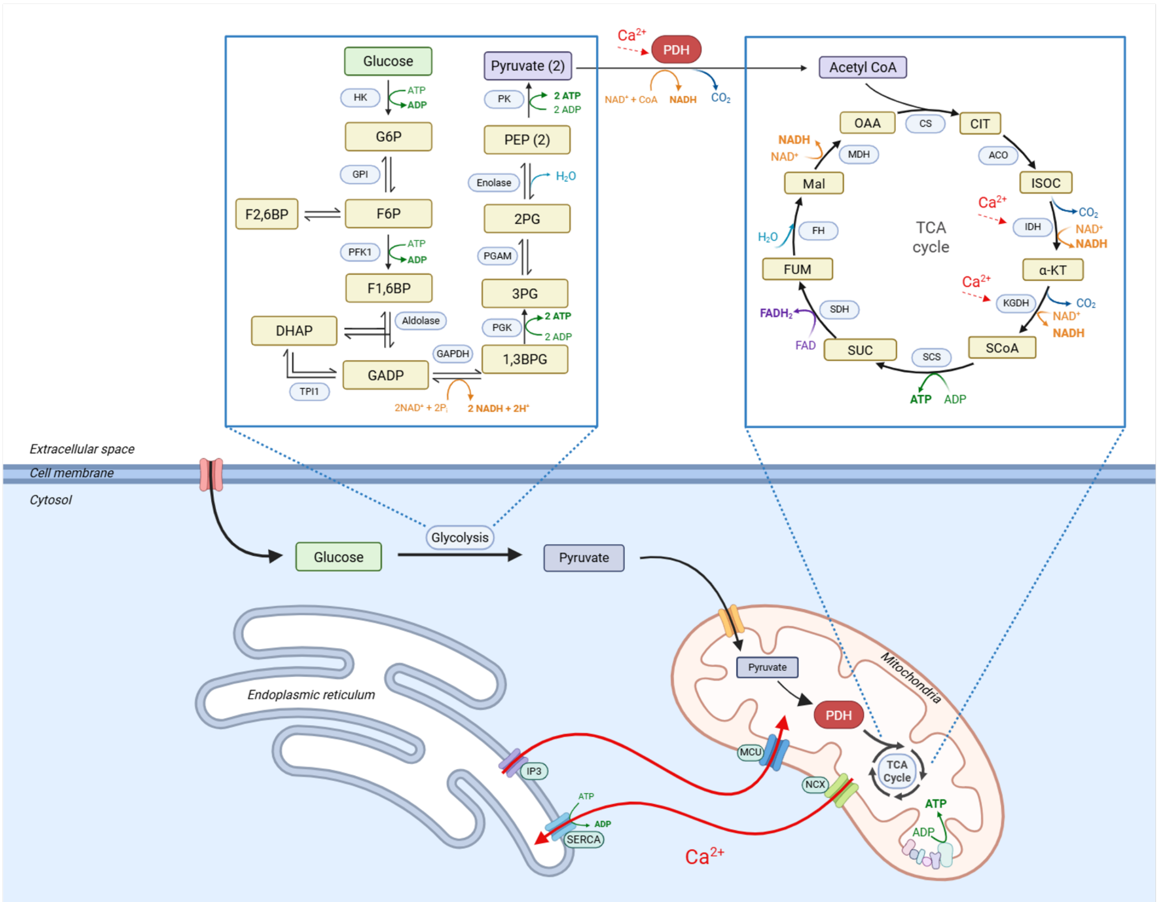


Figure 2: Biochemical pathways in astrocyte glucose metabolism showing glycolysis, mitochondrial oxidation, and calcium-dependent regulation.

2 Theoretical Framework

2.1 Research Objective

We address three questions:

1. How do different noise formulations (additive, multiplicative, state-dependent, jump) affect calcium oscillation statistics?
2. What is the appropriate probability distribution for inter-spike intervals, and what does this indicate about the underlying mechanism?
3. How does the system respond to parameter variations, and can noise affect survival under pathological conditions?

2.2 Molecular Origins of Stochasticity

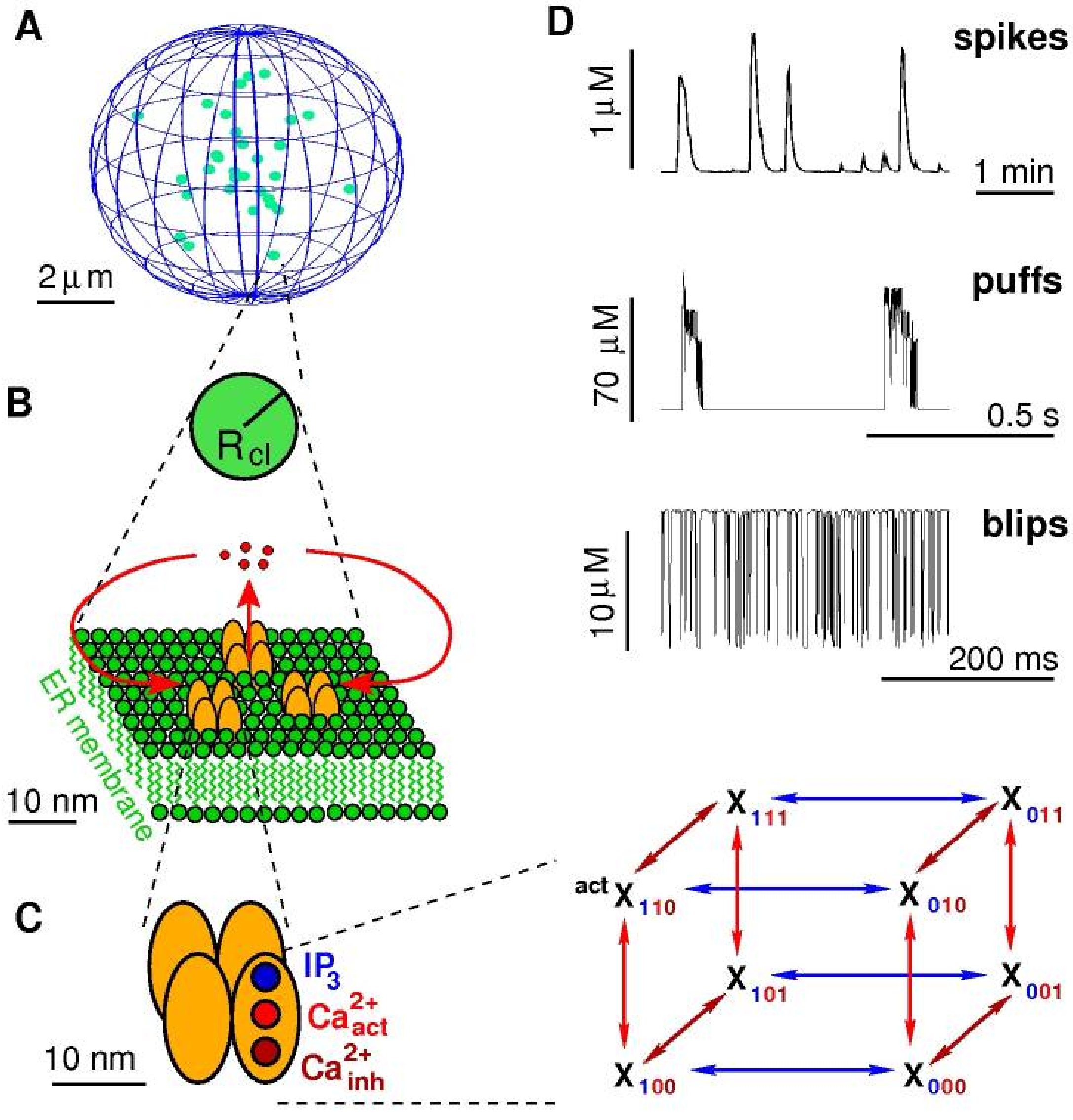


Figure 3: DeYoung-Keizer model for IP_3R channel gating, showing the eight-state Markov chain arising from three binding sites per subunit. From [11].

Stochasticity in calcium signaling arises from discrete molecular events. IP_3Rs are tetrameric channels where each subunit contains binding sites for IP_3 and two for Ca^{2+} (one activating, one inhibiting). The DeYoung-Keizer model represents each subunit as an 8-state Markov chain [13]. Channel opening requires specific binding configurations, with transition rates depending on local IP_3 and Ca^{2+} concentrations.

2.2.1 Hierarchical Signal Organization

Calcium signals exhibit hierarchical structure arising from spatial organization [10]:

Blips: Single channel openings release localized calcium (∼10–50 nm radius) lasting milliseconds.

Puffs: Channels cluster in groups of 5–20, separated by ∼1–7 μm. Calcium released by one channel diffuses to neighboring channels, triggering calcium-induced calcium release (CICR) within the cluster. This produces coordinated "puffs" involving the entire cluster.

Waves: If puff amplitude exceeds a threshold, calcium diffuses to adjacent clusters, potentially triggering cell-wide waves. The inter-cluster distance creates a natural filter: small puffs remain localized while large puffs can propagate.

This hierarchy creates scale-dependent equilibrium properties. At the channel level, fast Markov transitions (∼ms timescale) reach local quasi-equilibrium rapidly. At the cellular level, the system remains far from equilibrium, exhibiting excitable dynamics driven by channel noise [11].

2.2.2 Master Equation vs. Fokker-Planck Descriptions

For discrete Markov processes, the probability $P_n(t)$ of occupying state n evolves according to the Master Equation:

$$\frac{dP_n(t)}{dt} = \sum_{m \neq n} \left[W_{nm} P_m(t) - W_{mn} P_n(t) \right] \tag{1}$$

where W_{nm} denotes the transition rate from state m to n.

For continuous processes driven by Gaussian white noise, the Fokker-Planck equation governs probability density evolution:

$$\frac{\partial p(x,t)}{\partial t} = -\frac{\partial}{\partial x} \left[f(x) p(x,t) \right] + \frac{1}{2} \frac{\partial^2}{\partial x^2} \left[g^2(x) p(x,t) \right] \tag{2}$$

For calcium dynamics involving both discrete channel transitions and continuous concentration changes, Thurley and Falcke [12] derived integro-differential equations where future evolution depends on the history of past puff events. The continuous SDE formulations represent mean-field approximations might be inaccurate since it neglects discrete channel effects.

2.3 Deterministic Model

The deterministic model, developed by Voorsluijs et al. [14], comprises 30 coupled ODEs describing calcium dynamics, glycolysis, and mitochondrial metabolism. The core calcium equations are:

$$\begin{aligned}
\frac{d[\mathrm{Ca}^{2+}]_c}{dt} &= \mu_M f_c \left(-J_{\mathrm{SERCA}} + J_{\mathrm{ER,out}} + \delta (J_{\mathrm{NCX}} - J_{\mathrm{uni}}) \right) \\
\frac{d[\mathrm{Ca}^{2+}]_{\mathrm{ER}}}{dt} &= \mu_M \delta_{\mathrm{ER}} \left(J_{\mathrm{SERCA}} - J_{\mathrm{ER,out}} \right)
\end{aligned}$$

The IP3 receptor flux $J_{\mathrm{ER,out}}$ exhibits complex regulation:

$$J_{\mathrm{ER,out}} = \left[\left(v_{\mathrm{IP3}} \frac{[\mathrm{IP3}]^2}{[\mathrm{IP3}]^2 + K_{\mathrm{IP3}}^2} \frac{[\mathrm{Ca}^{2+}]_c^2}{[\mathrm{Ca}^{2+}]_c^2 + K_{\mathrm{act}}^2} \frac{K_{\mathrm{inh}}^4}{K_{\mathrm{inh}}^4 + [\mathrm{Ca}^{2+}]_c^4} \right) + v_{\mathrm{leak}} \right] \frac{[\mathrm{Ca}^{2+}]_{\mathrm{ER}} - [\mathrm{Ca}^{2+}]_c}{\mu_M}$$

This formulation encodes three regulatory mechanisms: IP_3 binding (activation), low calcium binding (activation), and high calcium binding (inhibition). The resulting bell-shaped calcium dependence generates oscillatory dynamics.

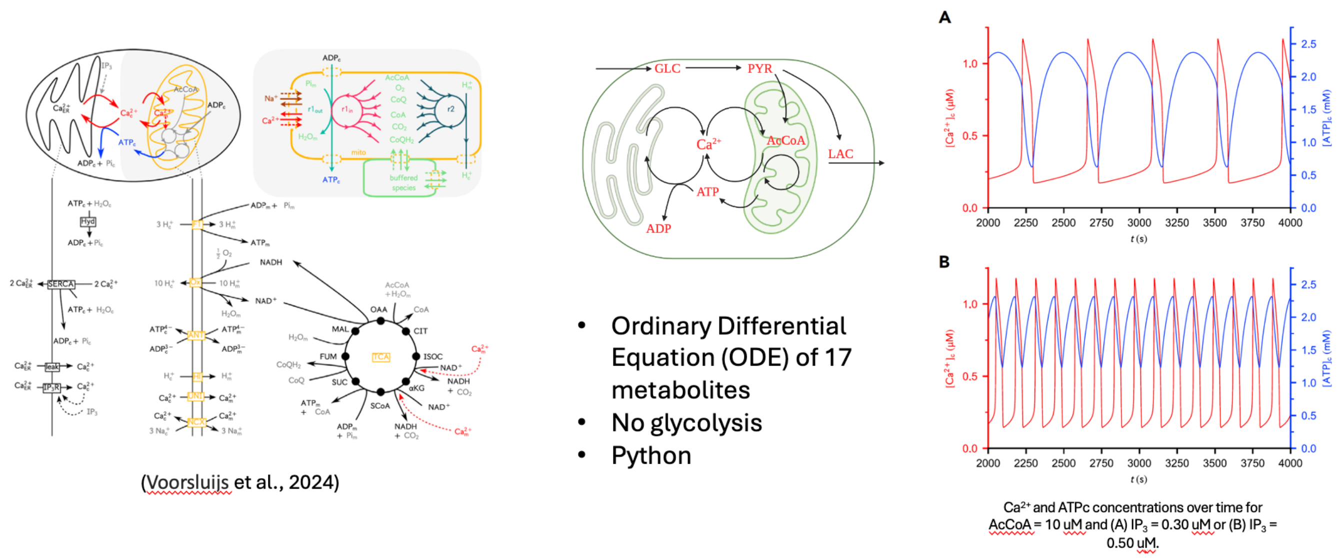


Figure 4: Model structure and typical dynamics. Left: interaction network. Center: simplified schematic. Right: predicted oscillations showing ∼100 s period.

2.4 Stochastic Perturbation Schemes

We introduce four noise formulations applied to the IP_3R calcium flux, each representing different physical mechanisms.

2.4.1 Additive Gaussian Noise

Additive noise represents perturbations independent of system state:

$$dX_t = f(X_t, t)dt + \sigma_{\text{add}} dW_t$$

where W_t is a standard Wiener process. For the calcium equations:

$$\begin{aligned} d[\text{Ca}^{2+}]_{\text{ER}} &= \mu_M \delta_{\text{ER}}(J_{\text{SERCA}} - J_{\text{ER,out}})dt - \sigma_{\text{add}}\mu_M\delta_{\text{ER}}dW_t \\ d[\text{Ca}^{2+}]_c &= \mu_M f_c(-J_{\text{SERCA}} + J_{\text{ER,out}} + \delta(J_{\text{NCX}} - J_{\text{uni}}))dt + \sigma_{\text{add}} f_c \delta_{\text{ER}} dW_t \end{aligned}$$

with $\sigma_{\text{add}} = 0.05$ (larger values cause numerical instability).

2.4.2 Multiplicative Noise

Multiplicative noise scales with flux magnitude:

$$dX_t = f(X_t, t)dt + \sigma_{\text{mult}}|J_{\text{ER,out}}|dW_t$$

with $\sigma_{\text{mult}} = 0.5$. This represents state-dependent channel fluctuations where larger fluxes produce proportionally larger noise.

2.4.3 State-Dependent Noise with Sublinear Scaling

Square-root scaling represents shot noise from discrete molecular events:

$$dX_t = f(X_t, t)dt + \sigma_{\text{state}}\sqrt{|J_{\text{ER,out}}|}dW_t$$

with $\sigma_{\text{state}} = 0.5$. The sublinear scaling reflects that relative fluctuations decrease with increasing flux magnitude, consistent with Poisson statistics for channel openings.

2.4.4 Jump Process Noise

Jump processes model discrete, intermittent perturbations:

$$dX_t = f(X_t, t)dt + \sigma_{\text{jump}} dN_t$$

where N_t is a Poisson process with rate $\lambda_{\text{jump}} = 0.01$ s^{-1} and jump magnitude $\sigma_{\text{jump}} = 0.01$ μM. This represents rare but significant events such as coordinated cluster openings.

3 Computational Methods

3.1 Implementation

Simulations were performed in Julia using DifferentialEquations.jl[1]. The deterministic system was solved using Tsit5 (explicit Runge-Kutta 5(4)). SDEs with continuous noise used ImplicitRKMil (implicit Runge-Kutta-Milstein) for stability with stiff dynamics. Jump processes used Rodas5P (Rosenbrock method) combined with direct Gillespie sampling for jump times.

Each simulation ran for 8000 s, with the first 4000 s discarded as transient. The final 4000 s (sampled at 1 Hz) were analyzed, capturing approximately 40 complete oscillation cycles. For noise strength scans, shorter simulations (4000 s total, 1000 s analyzed) maintained sufficient oscillation counts while managing computational cost.

3.2 Analysis Metrics

ATP:ADP Ratio: Computed as $R(t) = [\text{ATP}]_c(t)/[\text{ADP}]_c(t)$, serving as a proxy for cellular energetic state.

Peak Detection: Local maxima exceeding the 75th percentile of the time series were identified as oscillation peaks.

Inter-Spike Intervals (ISI): Time differences between consecutive peaks: $\text{ISI}_k = t_{k+1} - t_k$.

Shannon Entropy: Computed from the ISI histogram: $H = -\sum_i p_i \log_2 p_i$, where p_i is the probability of ISI falling in bin i.

Escape Rate: Defined as the inverse mean ISI: $\lambda_{\text{escape}} = 1/\langle \text{ISI} \rangle$, analogous to Kramers escape rate.

4 Results

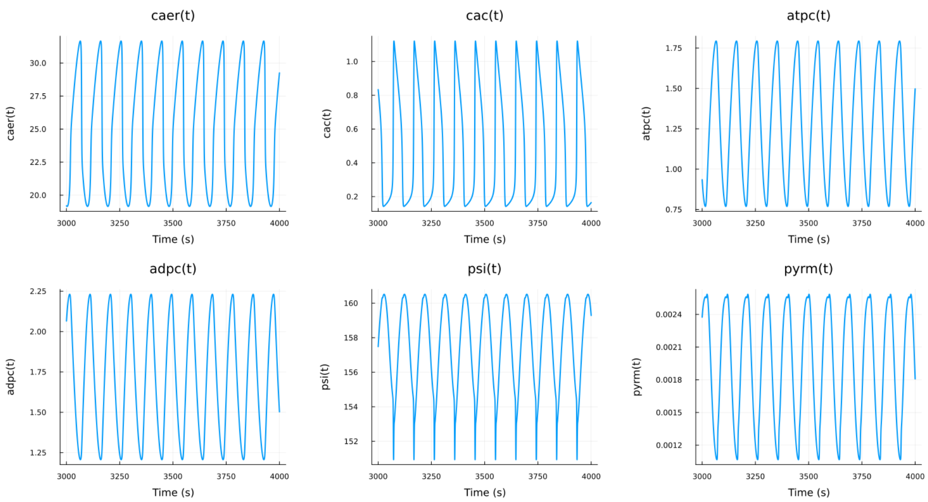


Figure 5: Deterministic dynamics showing six key metabolic variables.

[1]Code available at `https://github.com/bingyulab/SPDE`

Figure 5 shows deterministic dynamics at $IP_3 = 0.7$ μM. The system exhibits regular oscillations with period $\sim$100 s. Cytosolic calcium (CaC) and ATP oscillate in phase, while ER calcium (CaER) oscillates in antiphase—consistent with the release-refill mechanism. The ATP:ADP ratio cycles between 0.3 and 1.6.

4.1 Computational Performance

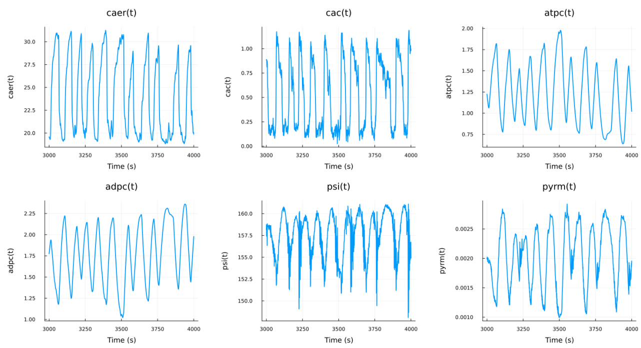
Table 1 summarizes computational performance and convergence properties across all noise conditions. All simulations successfully reached the target time of 8000 seconds.

Table 1: Simulation Statistics for IP3 = 0.7 μM

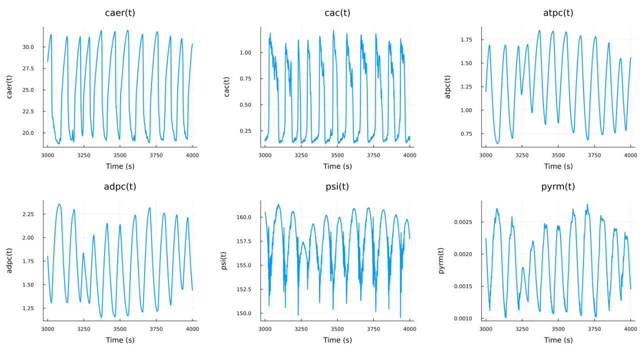
Noise Type	Algorighm	**Steps**	**Solve Time (s)**	**Status**
None (Deterministic)	Tsit5	24457	1.44	Success
Additive	ImplicitRKMil	4001	7.09	Success
Multiplicative	ImplicitRKMil	4001	9.20	Success
State-dependent	ImplicitRKMil	4001	9.63	Success
Jump	Rodas5P	6897	0.78	Success

4.2 Effects of Noise Type on Dynamics

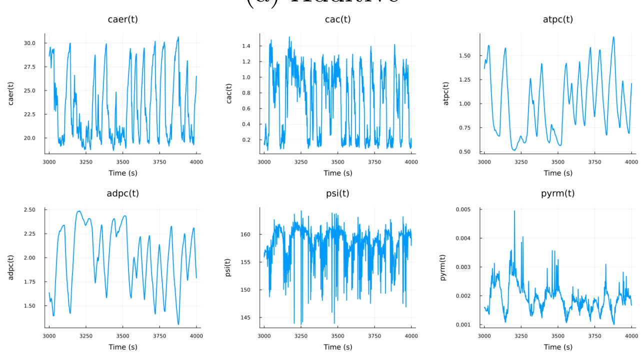
Figure 6 compares dynamics under different noise conditions. Additive, multiplicative, and jump noise preserve the basic oscillatory structure with minor amplitude variations. State-dependent noise produces qualitatively different behavior: irregular amplitude modulation creates a pattern where oscillation amplitude varies on timescales longer than the fundamental period.



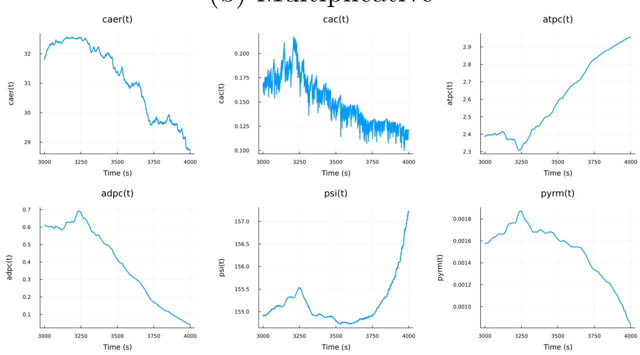
(a) Additive



(b) Multiplicative



(c) State-dependent

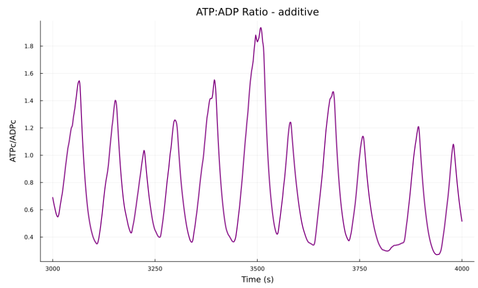


(d) Jump

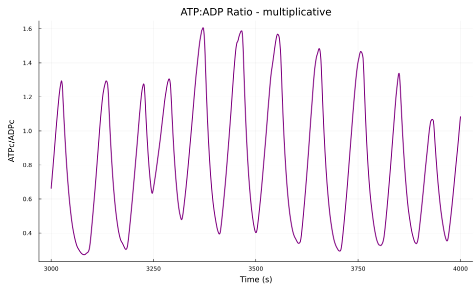
Figure 6: Key metabolic variables under different noise conditions at $IP_3 = 0.7$ μM.

4.3 ATP:ADP Ratio Analysis

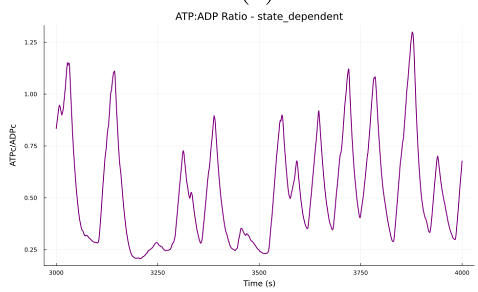
Figure 7 shows ATP:ADP ratio time series. Additive, multiplicative, and jump noise maintain regular oscillations. State-dependent noise produces higher variability with amplitude modulation on slow timescales.



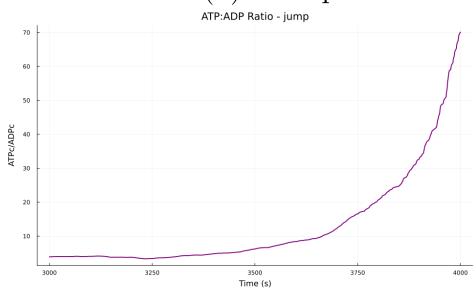
(a) Additive



(b) Multiplicative



(c) State-dependent

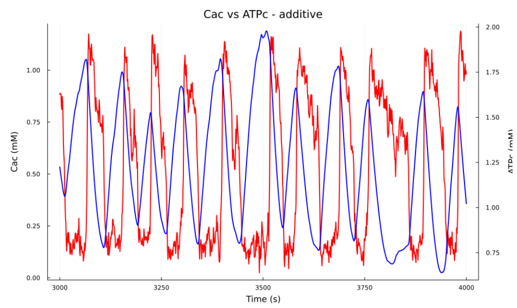


(d) Jump

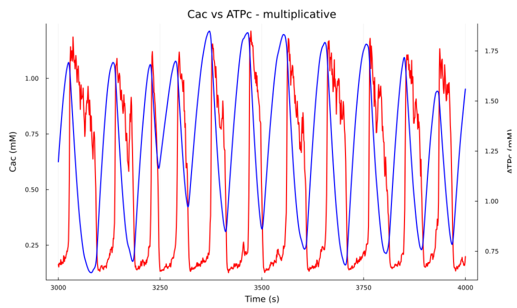
Figure 7: ATP:ADP ratio time evolution under different noise conditions.

4.4 Calcium-ATP Phase Relationships

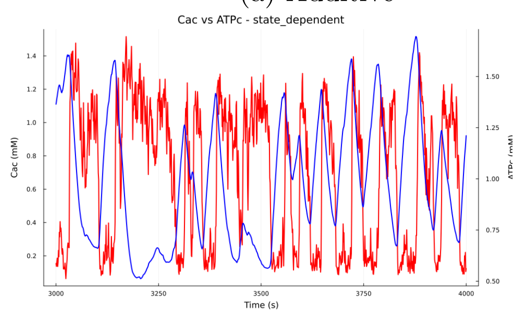
Figure 8 overlays cytosolic calcium and ATP time courses. All noise types preserve the phase relationship where ATP peaks follow calcium peaks. State-dependent noise disrupts peak alignment.



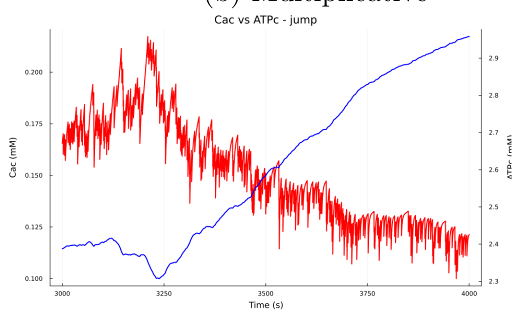
(a) Additive



(b) Multiplicative



(c) State-dependent

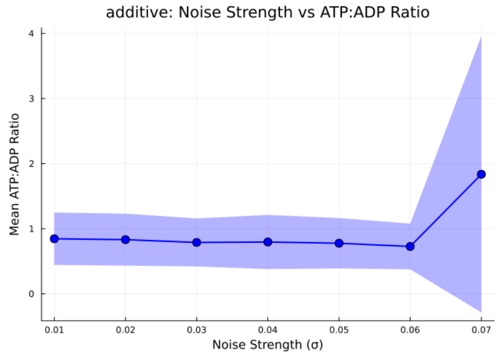


(d) Jump

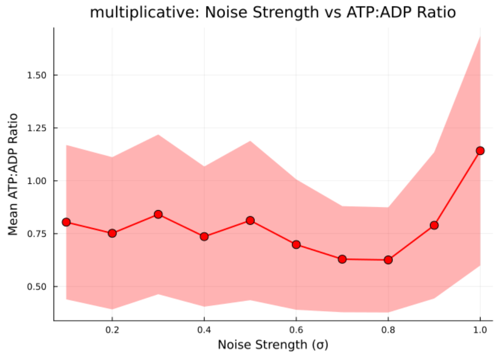
Figure 8: Overlaid CaC (blue, left axis) and ATP (red, right axis) time courses.

4.5 Noise Strength Dependence

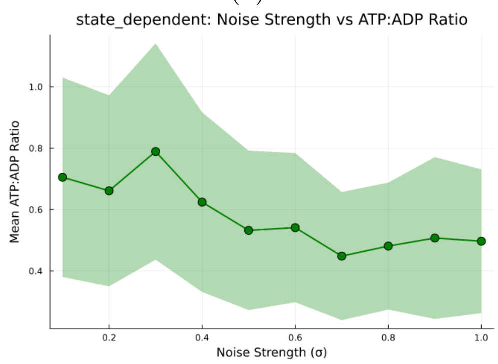
Figure 9 shows mean ATP:ADP ratio versus noise strength, averaged over 10 independent simulations. Additive noise has minimal effect across the tested range (0.01–0.07), consistent with effective filtering of constant-amplitude perturbations. Multiplicative noise shows monotonic increase with noise strength, reflecting enhanced activation during high-flux states. State-dependent noise shows non-monotonic response: initial increase followed by decline at high noise. Jump noise maintains near-constant ratio until threshold, then shows abrupt transition.



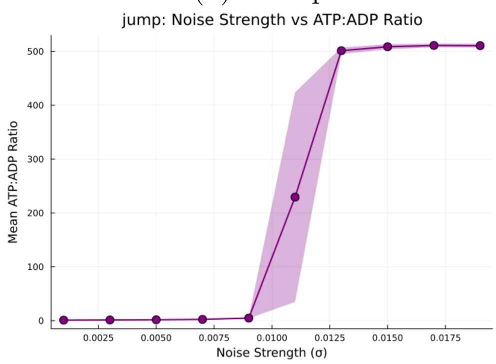
(a) Additive



(b) Multiplicative



(c) State-dependent

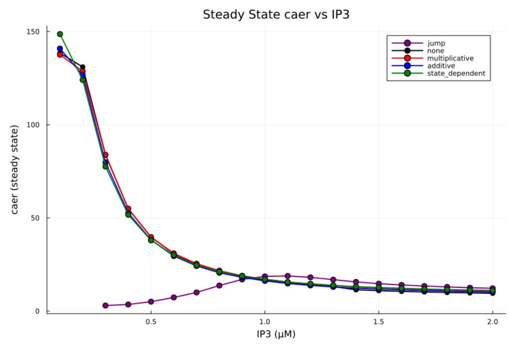


(d) Jump

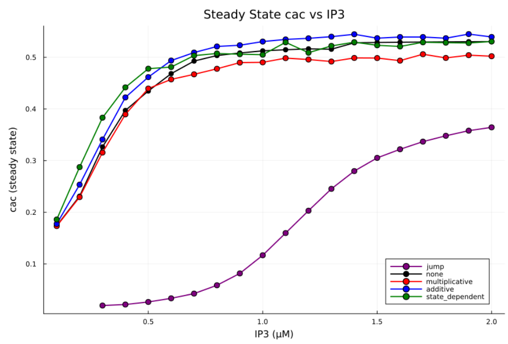
Figure 9: Mean ATP:ADP ratio versus noise strength (10-run ensemble average ± SD).

4.6 IP3 Parameter Scan

Figures 10 shows steady-state metabolite concentrations across IP_3 values from 0.1 to 2.0 μM. ER calcium decreases monotonically with IP_3, consistent with increased release through IP_3Rs.



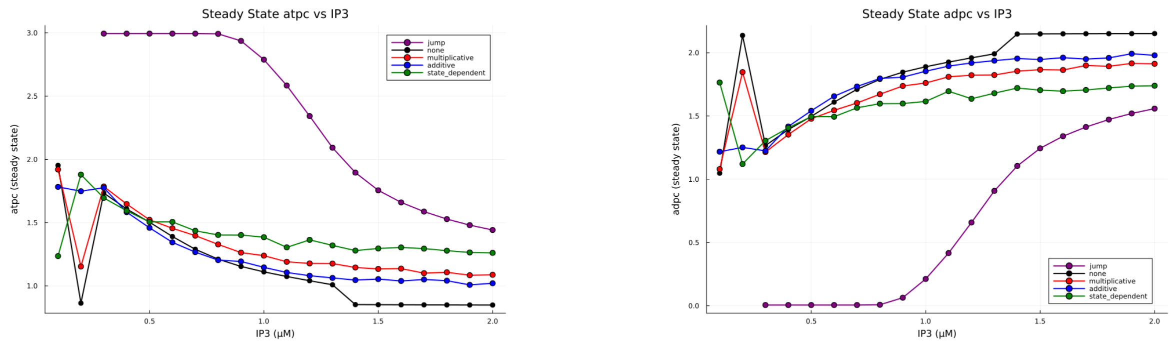
(a) ER Calcium vs IP3



(b) Cytosolic Calcium vs IP3

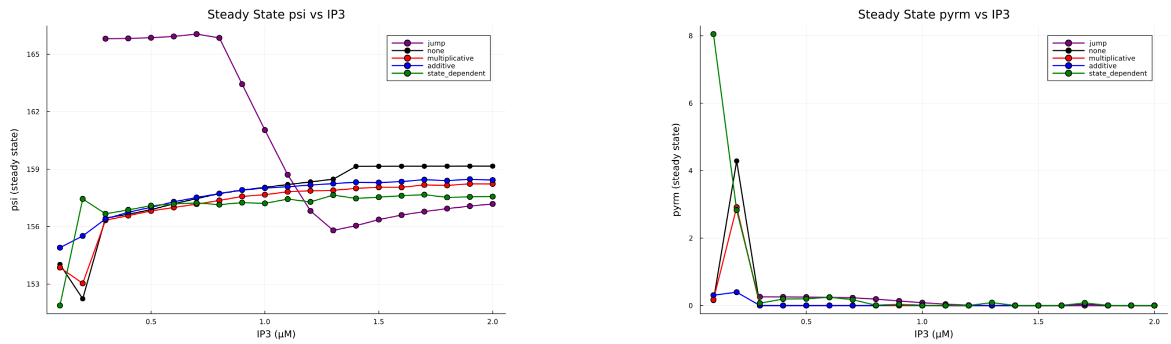
Figure 10: Steady-state calcium concentrations versus IP_3.

Figures 11 12 shows biphasic response: initial increase (more release) followed by plateau (depletion of ER stores). ATP follows a similar pattern to cytosolic calcium, reflecting the calcium-dependent stimulation of mitochondrial metabolism.



(a) Cytosolic ATP vs IP3 (b) Cytosolic ADP vs IP3

Figure 11: Steady-state cytosolic ATP and ADP versus IP_3.



(a) Mitochondrial Membrane Potential vs IP3 (b) Mitochondrial Pyruvate vs IP3

Figure 12: Steady-state mitochondrial concentrations versus IP_3 (10-run ensemble average).

4.7 Spectral Analysis

Figure 13 shows frequency variance across IP_3 values, quantifying oscillation regularity. Low variance indicates deterministic-like regularity; high variance indicates irregular dynamics. Additive, multiplicative, and state-dependent noise maintain moderate variance across most IP_3 values. Jump noise shows high variance at low IP_3 (irregular, noise-dominated) transitioning to low variance near $IP_3 = 1.5$ μM (regular oscillations).

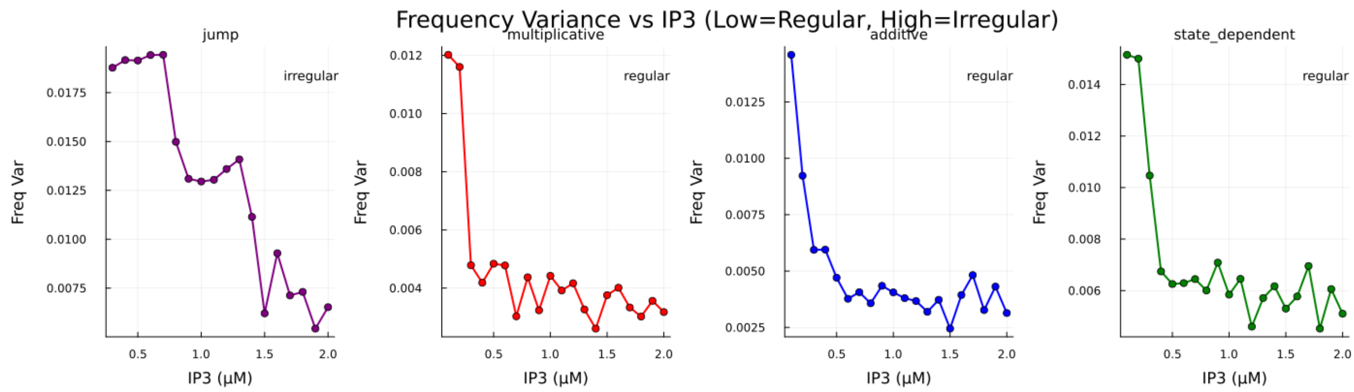
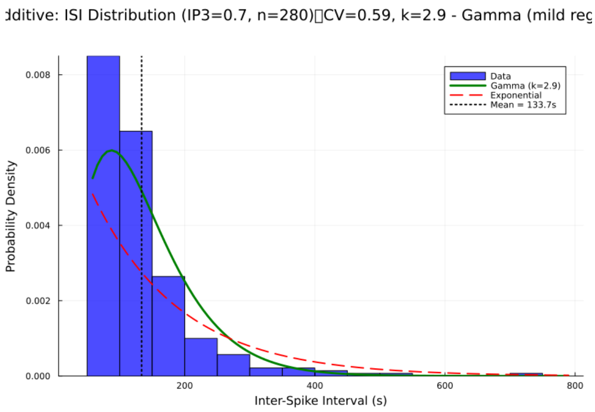


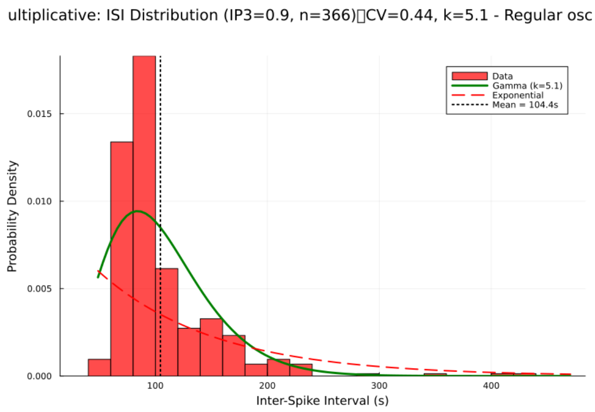
Figure 13: Dominant Frequency variance vs IP_3 (lower variance = more regular)

4.8 Inter-Spike Interval Statistics

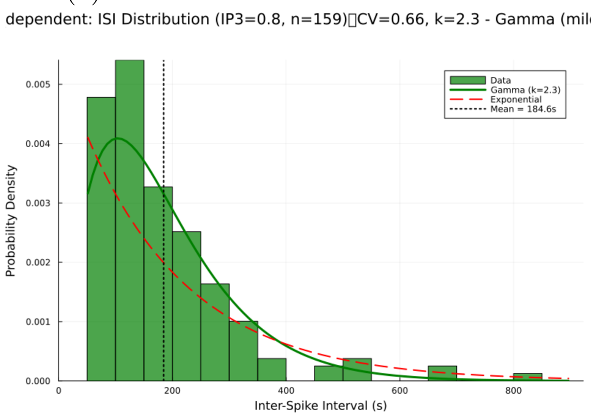
Figure 14 shows ISI distributions at $IP_3 = 0.7\ \mu M$ with exponential (Kramers) and Gamma distribution fits. The Gamma distribution provides superior fits for all noise types, with shape parameters $k > 1$ indicating the distributions have modes away from zero, which is inconsistent with simple Poisson (exponential) statistics.



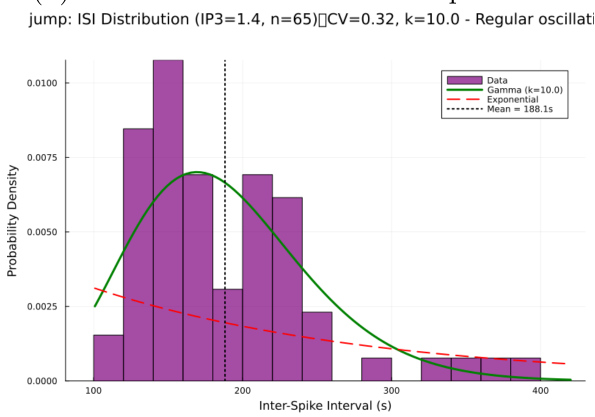
(a) The ISI distribution of Additive Noise



(b) The ISI distribution of Multiplicative Noise



(c) The ISI distribution of State-Dependent Noise



(d) The ISI distribution of Jump Noise

Figure 14: ISI distributions with exponential (red dashed) and Gamma (green solid) fits.

The Gamma distribution has probability density:

$$p(\tau; k, \theta) = \frac{\tau^{k-1} e^{-\tau/\theta}}{\Gamma(k)\theta^k} \tag{3}$$

For $k = 1$, this reduces to the exponential distribution predicted by Kramers theory. The observed $k > 1$ indicates that oscillation generation involves multiple sequential steps rather than single barrier crossing. Biologically, this reflects the requirement for ER refilling, IP_3R recovery from calcium-dependent inactivation, and threshold crossing—processes that must complete sequentially before the next oscillation can occur.

The coefficient of variation for Gamma distributions is $CV = 1/\sqrt{k}$. Observed values of $k \approx 3$–10 correspond to $CV \approx 0.3$–0.6.

4.9 Kramers Analysis

Figure 15 examines the relationship between ISI entropy and ATP:ADP ratio (as an energy proxy). Classical Kramers theory predicts that higher energy barriers produce longer escape times and potentially higher entropy. However, we observe negative correlations: higher ATP:ADP corresponds to lower entropy (more regular oscillations). **This suggested that kramers theory does not**

predict entropy production or regularity of oscillations, it predicts mean escape rates only.

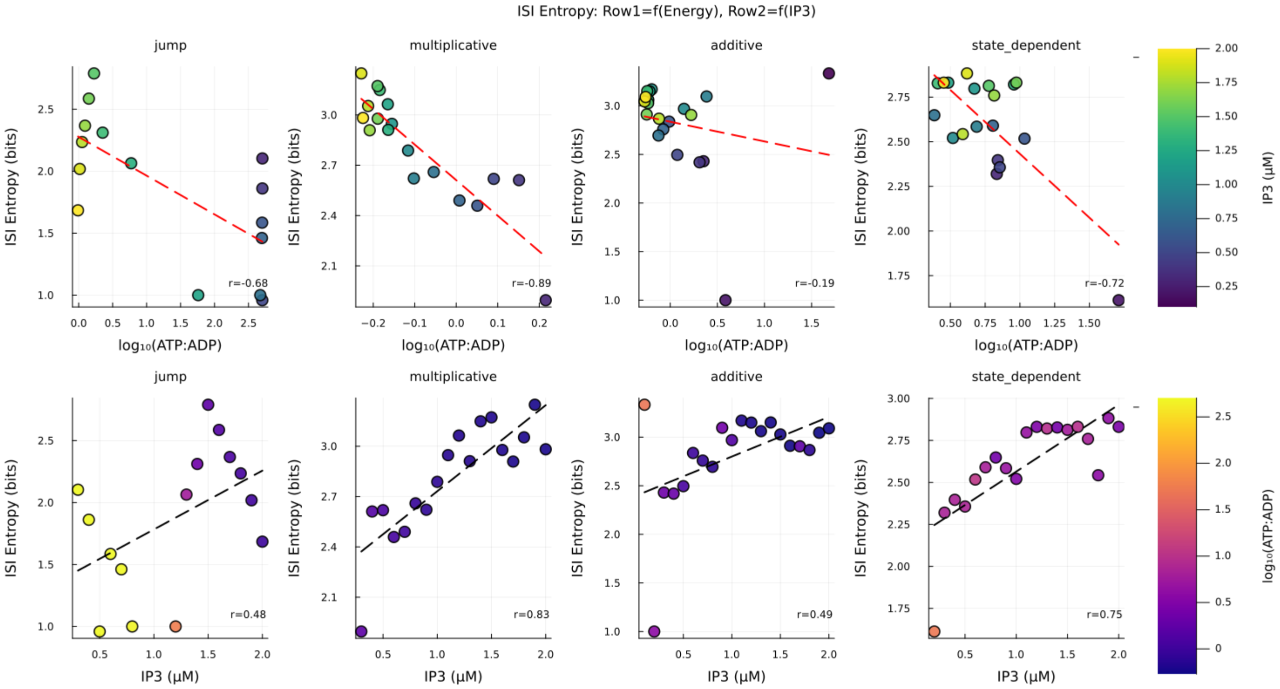


Figure 15: ISI entropy versus ATP:ADP ratio for all noise types. Points are colored by IP_3 value. Dashed lines show linear regression.

Figure 16 shows escape rate versus log(ATP:ADP ratio). Multiplicative, state-dependent, and jump noise show negative correlations consistent with Kramers-like behavior: higher "energy" corresponds to lower escape rates. Additive noise shows positive correlation—an "anti-Kramers" phenomenon where higher energy corresponds to faster escape.

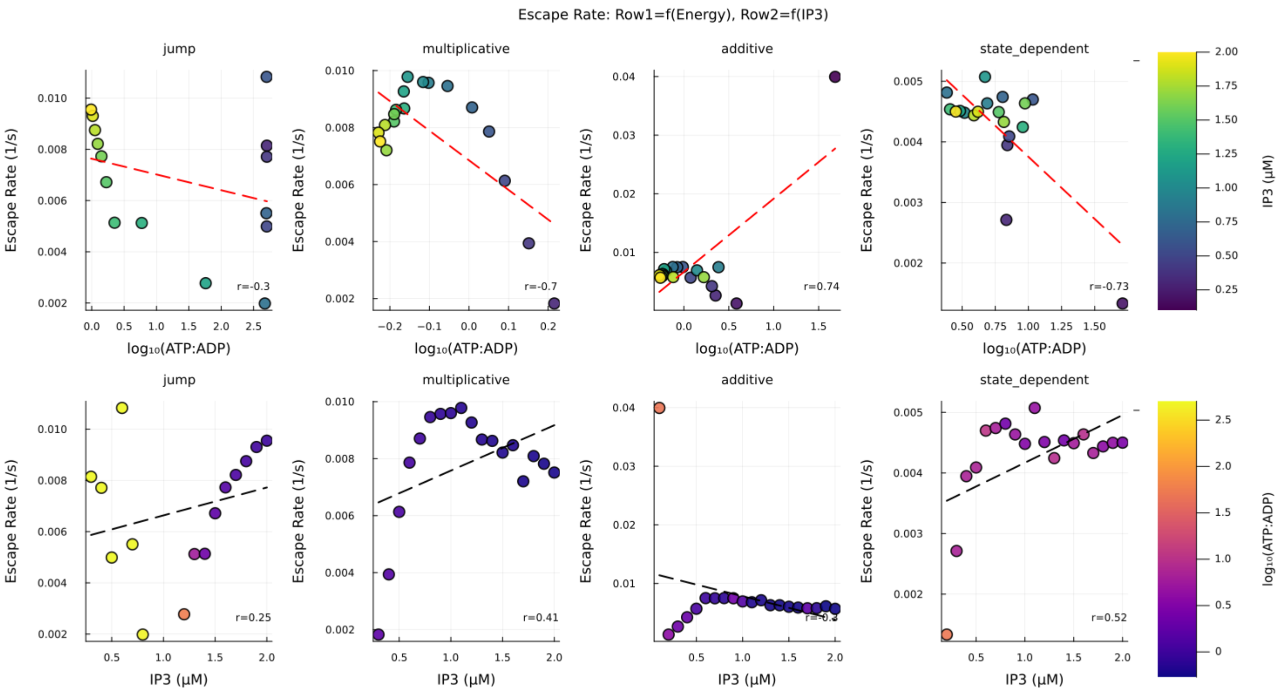


Figure 16: Escape rate versus log(ATP:ADP ratio). Points colored by IP_3 value.

Table 2 summarizes correlation coefficients. The sign of log(Energy)-log(Rate) correlation distinguishes noise types: negative correlations (multiplicative, state-dependent, jump) indicate Kramers-like dynamics; positive correlation (additive) indicates anti-Kramers behavior.

Table 2: Kramers Correlation Summary

Noise Type	logE-Rate	logE-logR	IP3-Rate	Kramers?
jump	-0.305	-0.345	0.251	Yes (E↑→R↓)
multiplicative	-0.699	-0.721	0.413	Yes (E↑→R↓)
additive	0.738	0.346	-0.302	No (Anti-Kramers)
state_dependent	-0.725	-0.759	0.517	Yes (E↑→R↓)

The anti-Kramers behavior of additive noise can be understood as follows. Additive noise has constant amplitude regardless of system state. When ATP:ADP is high, the system executes large-amplitude oscillations. The constant noise amplitude becomes relatively small compared to oscillation amplitude, reducing its ability to perturb the cycle. In contrast, state-dependent noise scales with flux magnitude, coupling directly to the oscillation amplitude and producing classical Kramers-like behavior.

4.10 Bifurcation Analysis

Figure 17 characterizes the deterministic bifurcation structure. At low IP_3 (< 0.3 μM), the system resides in a stable fixed point with no oscillations. Near $IP_3 = 0.3$ μM, a supercritical Hopf bifurcation gives birth to oscillations with initially high ATP:ADP ratio. As IP_3 increases, oscillation amplitude peaks near 0.4 μM then decreases. The oscillation frequency increases with IP_3 until approximately 1.3 μM.

Near $IP_3 = 1.4$ μM, a second bifurcation occurs: oscillations cease and the system transitions to a pathological fixed point with ATP:ADP ≈ 0. This is not merely loss of oscillations but metabolic collapse—ATP production effectively stops.

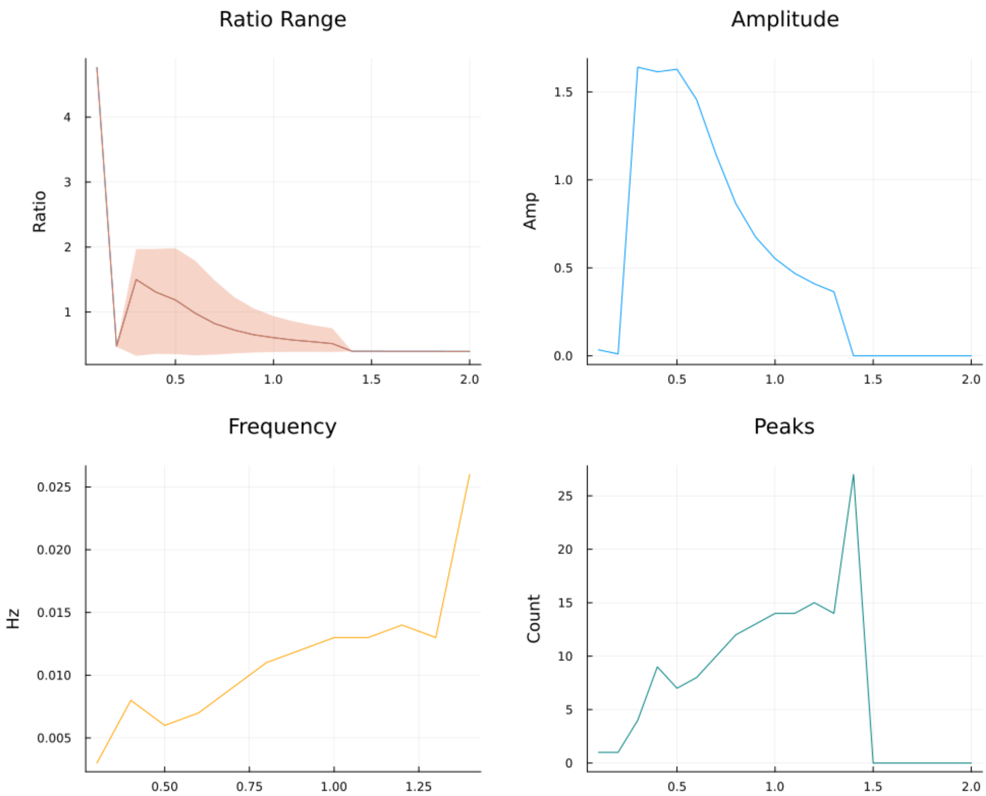


Figure 17: Bifurcation summary showing ATP:ADP ratio (with min-max envelope), oscillation amplitude, dominant frequency, and peak count versus IP_3.

Figure 18 compares time series at three IP_3 values spanning the upper bifurcation. At $IP_3 = 1.2\ \mu M$ (below bifurcation), regular oscillations persist. At $IP_3 = 1.4\ \mu M$ (near bifurcation), oscillations have nearly ceased. At $IP_3 = 1.6\ \mu M$ (above bifurcation), the system has collapsed to a fixed point with near-zero ATP.

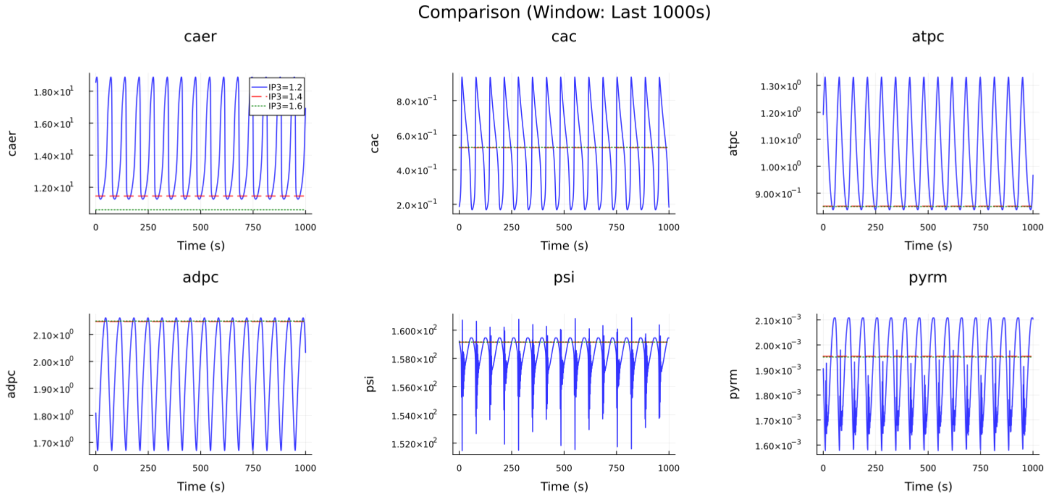
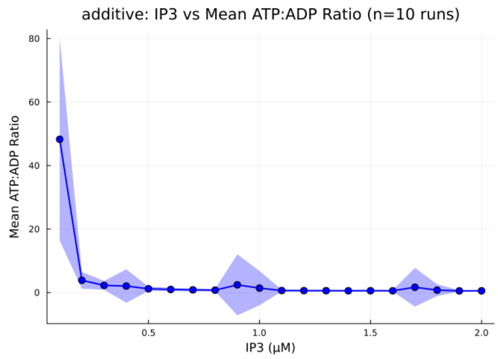


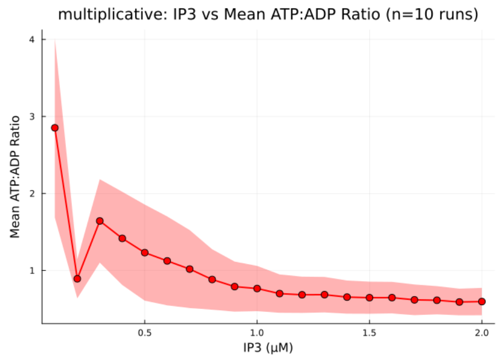
Figure 18: Time series of six key variables comparing $IP_3 = 1.2\ \mu M$ (below bifurcation, blue), $IP_3 = 1.4\ \mu M$ (near bifurcation, red), and $IP_3 = 1.6\ \mu M$ (after bifurcation, green).

4.11 Noise-Dependent Survival at High IP_3

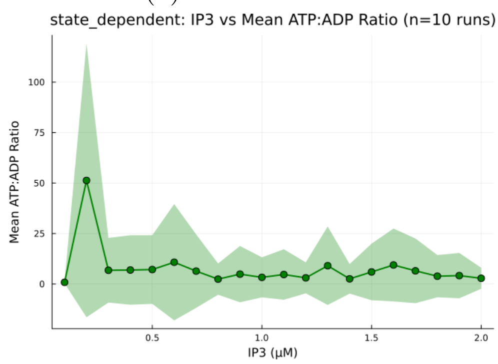
Figure 19 reveals that noise type determines whether cells survive or die at high IP_3. The deterministic system and additive noise both collapse to ATP:ADP ≈ 0 above $IP_3 = 1.4\ \mu M$. Jump noise shows dramatic collapse from ratio $\sim$500 to zero around $IP_3 = 1.2\ \mu M$.



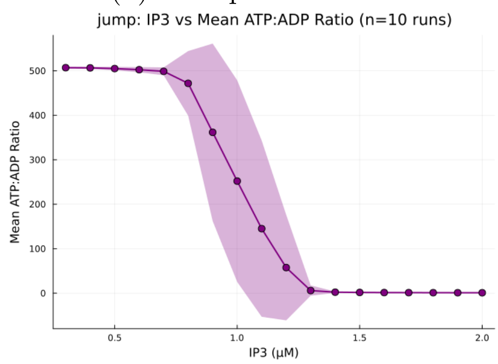
(a) Additive Noise vs IP3



(b) Multiplicative Noise vs IP3



(c) State-Dependent Noise vs IP3



(d) Jump Noise vs IP3

Figure 19: ATP:ADP ratio versus IP_3 for each noise type (10-run ensemble average $\pm$ SD).

In contrast, multiplicative noise maintains ATP:ADP ≈ 0.6 even at $IP_3 = 2.0$ μM—reduced but positive. State-dependent noise maintains ATP:ADP $\approx$ 5–20 with high variability throughout the tested range. These noise types provide "stochastic rescue": random perturbations prevent the system from settling into the pathological attractor.

5 Discussion

5.1 Hierarchical Noise Filtering

The calcium-metabolic system exhibits hierarchical noise filtering at moderate parameter values. Perturbations to IP_3R flux are absorbed primarily by the ER calcium compartment, which acts as a buffer. The ATP:ADP ratio, located downstream in the metabolic cascade, remains relatively stable. This architectural feature may represent an evolutionary adaptation: metabolic output should remain reliable despite inevitable molecular fluctuations in signaling components.

5.2 Gamma Distributions and Multi-Step Oscillation Mechanisms

The superior fit of Gamma distributions over exponential distributions has mechanistic implications. The shape parameter $k > 1$ indicates that oscillation generation requires multiple sequential steps to complete. Biologically, this reflects:

1. ER calcium depletion via IP_3R opening
2. SERCA-mediated ER refilling (ATP-dependent)
3. IP_3R recovery from calcium-dependent inactivation
4. Threshold crossing for the next release event

The Gamma distribution naturally describes the waiting time for k sequential exponential steps. This connection between statistical analysis and biochemical mechanism validates the modeling approach.

5.3 Anti-Kramers Phenomenon

The positive energy-rate correlation for additive noise contradicts classical Kramers theory but has a straightforward explanation. Kramers theory assumes noise helps the system escape from energy minima by providing random kicks. For additive noise with constant amplitude, this mechanism operates uniformly regardless of system state.

However, when ATP:ADP is high, the system executes large-amplitude oscillations. The constant noise amplitude becomes relatively insignificant compared to the deterministic dynamics, so oscillation completion is dominated by intrinsic dynamics rather than noise-assisted escape. The observed positive correlation reflects that high-energy states (high ATP:ADP) correspond to fast, robust oscillations that complete quickly regardless of noise.

For state-dependent and multiplicative noise, the noise amplitude scales with system state, maintaining coupling to the oscillation dynamics across the entire parameter range. This produces the negative energy-rate correlation predicted by Kramers theory.

5.4 Stochastic Rescue and Cell Survival

The observation that state-dependent and multiplicative noise maintain positive ATP production at high IP_3 while additive noise permits metabolic collapse represents a form of noise-induced stabilization. When the deterministic system would settle into a pathological fixed point (ATP:ADP = 0), appropriately structured noise prevents this by continuously perturbing the system away from the attractor.

The mechanism differs between noise types:

- **State-dependent noise:** Square-root scaling means that noise amplitude decreases as flux decreases. However, even small random perturbations can trigger transient calcium releases that prevent complete shutdown.
- **Multiplicative noise:** Linear scaling maintains stronger coupling to system state, providing continuous perturbations that prevent convergence to the pathological fixed point.
- **Additive noise:** Constant amplitude provides no state-dependent correction. When the deterministic dynamics drive toward the pathological state, constant noise cannot provide targeted rescue.
- **Jump noise:** Discrete events are too infrequent ($\lambda = 0.01$ s^{-1}) to prevent continuous decline between jumps.

5.5 Implications for Parkinson's Disease

These findings suggest several connections to PD pathophysiology:

Bifurcation and Cell Death: The bifurcation at $IP_3 \approx 1.4$ μM, where ATP:ADP collapses to zero, represents metabolic failure consistent with the calcium overload hypothesis of neurodegeneration [15]. Excessive calcium release depletes ER stores, overwhelms mitochondrial buffering capacity, and triggers energy crisis.

Noise Type and Disease Progression: In PD, mitochondrial dysfunction creates variable responses to calcium signals—a form of state-dependent stochasticity. Our results suggest this variability might paradoxically extend survival: neurons persist in a dysfunctional-but-alive state with irregular metabolism rather than undergoing rapid collapse. This could explain the long prodromal phase of PD before clinical symptoms emerge.

Safety Window: The oscillatory regime ($IP_3 = 0.3$–1.3 μM) represents a "safety window" for normal function. Disease processes that shift neurons toward either boundary (reduced IP_3 signaling from receptor damage, or excessive calcium dysregulation) could trigger dysfunction or death.

5.6 Limitations

Several limitations should be noted. First, we examined only four noise formulations, all based on Gaussian or Poisson statistics. Biological systems may exhibit heavier-tailed fluctuations or long-range correlations (e.g., fractional Brownian motion) not captured here. Second, the mean-field SDE approach averages over many channels, potentially missing phenomena that emerge from explicit channel dynamics. Third, spatial aspects (calcium diffusion, organelle positioning) were not modeled. Fourth, the model describes a single astrocyte without considering network effects or neuron-astrocyte interactions.

6 Conclusions

This study establishes several principles regarding stochastic perturbations in calcium-metabolic systems:

1. **Hierarchical filtering:** The system exhibits noise filtering where ER calcium absorbs perturbations while downstream metabolic outputs remain stable.
2. **Multi-step oscillation mechanism:** ISI distributions follow Gamma rather than exponential statistics, indicating that oscillations require multiple sequential biochemical steps.

3. **Noise-type specificity:** Different noise formulations produce qualitatively different dynamics. State-dependent noise uniquely disrupts regularity through amplitude modulation. Additive noise shows anti-Kramers behavior with positive energy-rate correlation.

4. **Bifurcation structure:** The deterministic system exhibits two bifurcations: oscillation onset near $IP_3 = 0.3$ μM and metabolic collapse near $IP_3 = 1.4$ μM.

5. **Stochastic rescue:** At high IP_3 where deterministic dynamics collapse, state-dependent and multiplicative noise maintain positive ATP production through continuous perturbation. Additive and jump noise fail to provide this rescue.

6. **Disease implications:** The results suggest that mitochondrial dysfunction in PD may create state-dependent variability that paradoxically extends survival at the cost of metabolic irregularity, potentially explaining the long prodromal disease phase.

Future work should explore rougher noise types (fractional Brownian motion, Lévy processes), explicit channel dynamics, and network effects in astrocyte populations.

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A Numerical Methods for dynamics Systems

A.1 Tsit5 for Non-stiff Systems

The Tsitouras 5/4 **Runge-Kutta method** (Tsit5) is an explicit adaptive step-size integrator designed for non-stiff ordinary differential equations. It belongs to the family of Runge-Kutta pairs, specifically providing a fifth-order solution for the propagation step and a fourth-order solution for error estimation.

The method computes the next state y_{n+1} using a weighted sum of seven intermediate stages k_i:

$$y_{n+1} = y_n + h \sum_{i=1}^{7} b_i k_i$$

where each stage k_i evaluates the derivative function $f(t, y)$ at specific points within the interval $[t_n, t_{n+1}]$.

Tsit5 is optimized to minimize the norm of the principal truncation error coefficients, making it generally more efficient than the classic Dormand-Prince (DoPri5) method often used as a default in other software packages (like MATLAB's `ode45`). It features the "First Same As Last" (FSAL) property, meaning the last stage evaluation of step n is identical to the first stage of step $n + 1$, effectively reducing the cost to 6 function evaluations per step despite having 7 stages.

For non-stiff problems where the timescales of dynamics are relatively uniform, Tsit5 offers an excellent balance of accuracy and computational speed, making it the default choice in the Julia DifferentialEquations.jl ecosystem for such systems.

A.2 Evolution of Stochastic Solvers

The simplest approach to stochastic differential equations (SDEs) is the **Euler-Maruyama method**, which extends the classical Euler method for ordinary differential equations:

$$X_{n+1} = X_n + f(X_n, t_n)\Delta t + g(X_n, t_n)\Delta W_n$$

where ΔW_n represents a random increment drawn from $\mathcal{N}(0, \Delta t)$. This method has strong convergence order 0.5 and weak convergence order 1.0, meaning it captures statistical properties but individual trajectories converge slowly. For systems requiring high accuracy or those with multiplicative noise, the Euler-Maruyama method becomes computationally expensive due to the need for very small time steps.

The **Milstein method** improves upon Euler-Maruyama by including an additional correction term that accounts for the interaction between the noise and its gradient:

$$X_{n+1} = X_n + f(X_n, t_n)\Delta t + g(X_n, t_n)\Delta W_n + \frac{1}{2} g(X_n, t_n) \frac{\partial g}{\partial x}(X_n, t_n)[(\Delta W_n)^2 - \Delta t]$$

This achieves strong convergence order 1.0 for scalar SDEs, significantly improving accuracy. However, both Euler-Maruyama and Milstein are explicit methods, which suffer from severe stability restrictions when applied to stiff systems—those containing processes operating on vastly different time scales.

In our calcium-metabolic system, fast calcium fluxes (millisecond scale) interact with slow metabolic processes (second to minute scale), creating stiffness. Explicit methods require time steps determined by the fastest process, even when we are primarily interested in slower dynamics. This motivates implicit methods, which solve nonlinear equations at each step to achieve stability.

A.3 Implicit Runge-Kutta-Milstein (ImplicitRKMil)

The ImplicitRKMil method combines three key advances: implicit time-stepping for stability, Runge-Kutta stages for high-order deterministic accuracy, and Milstein corrections for stochastic accuracy. The general form involves solving:

$$X_{n+1} = X_n + \sum_{i=1}^{s} b_i k_i + g(X_n, t_n)\Delta W_n + \frac{1}{2} g(X_n, t_n) \frac{\partial g}{\partial x}(X_n, t_n)[(\Delta W_n)^2 - \Delta t]$$

where the stages k_i are computed implicitly through coupled nonlinear equations:

$$k_i = f\left(X_n + \sum_{j=1}^{s} a_{ij} k_j, t_n + c_i \Delta t\right)\Delta t$$

The coefficients a_{ij}, b_i, and c_i are chosen to optimize both deterministic and stochastic convergence properties. The implicit equations are typically solved via Newton iteration at each time step. While computationally more expensive per step, the method permits much larger time steps for stiff systems, yielding overall efficiency gains.

For our additive, multiplicative, and state-dependent noise simulations, ImplicitRKMil provides stable integration with adaptive time-stepping that automatically adjusts Δt based on local error estimates. This ensures accuracy during rapid calcium transients while maintaining efficiency during slow metabolic phases.

A.4 Rodas5P for Jump Processes

For the jump process simulations, we employ the Rodas5P solver, a fifth-order **Rosenbrock method** specifically optimized for stiff ordinary differential equations. Unlike fully implicit Runge-Kutta methods that require solving nonlinear equations via Newton iterations at every step, Rosenbrock methods are linearly implicit. They linearize the problem around the current solution, reducing the computational burden to solving a system of linear equations:

$$(I - \gamma \Delta t J)\, k_i = f\left(X_n + \sum_{j=1}^{i-1} \alpha_{ij} k_j\right) + \Delta t J \sum_{j=1}^{i-1} \gamma_{ij} k_j$$

where $J = \partial f / \partial x$ denotes the Jacobian matrix. This approach provides the necessary stability for stiff systems while maintaining computational efficiency.

Rodas5P features adaptive step-size control with embedded error estimation and is "stiffly accurate," meaning it retains high accuracy even for differential-algebraic equations or very stiff transients. This makes it ideal for the calcium oscillation model, where the solver must accurately resolve sharp, rapid transitions in calcium concentration interspersed with smoother metabolic evolution, while also accommodating the discrete discontinuities introduced by the jump process.